hello abstract

1 Setup and calibration

Let $\mathscr{H} = \{h_1, \ldots, h_N\}$ be the discrete set of candidate classes. Our aim is to build a function C that maps an input x to a subset $C(x) \subseteq \mathscr{H}$ of classes, such that the true class lies in the set with probability at least $1-\alpha$. Our input is described by scores from several predictors. For each candidate class h we have a raw score vector $s_h = (s_{h,1}, \ldots, s_{h,K_h}) \in ([0,1] \cup \{\mathrm{NA}\})^{K_h}$, where $s_{h,k}$ is the evidence of the k-th predictor in favour of class h, while NA means that this predictor gave no score for the given class. We do not impute missing scores (NA). Instead, we record the set of observed coordinates $m_h = \{k : s_{h,k} \neq \mathrm{NA}\}$ and define all scoring functions on the pairs (s_h, m_h) and not on s_h alone. A score close to 1 means strong evidence for h, thus we map every observed coordinate to the nonconformity score $v_{h,k} = 1 - s_{h,k}$. When h is the true class, the vector v_h concentrates near the origin, while if h is not the true class, the vector v_h is further away. In Section 2 we write v for such a vector v_h and drop the index h.

Suppose we have a labelled calibration sample $\{(x_i, y_i)\}$. The predictors that produce the scores are trained on data that is disjoint from this sample and the test points. For each class h, the points with $y_i = h$ are randomly split into the shape discovery set $S_{1,h}$ and the size scaling set $S_{2,h}$. Each pair $(v_h(x_i), m_h(x_i))$ shows the score pattern for class h when it is actually the true class. A separate envelope is fitted for every class, using only the samples of that class.

Using $S_{1,h}$, we construct the scalar score $\tau_h(v,m)$, using one of the three envelope construction techniques in Section 2. We then compute the class specific scale factor $\hat{t}_h$ on $S_{2,h}$ as:

$$\hat{t}_h = \tau_{h,(k_h)}, \qquad k_h = \lceil (|S_{2,h}| + 1)(1-\alpha) \rceil, \tag{1}$$

where $\tau_{h,(k)}$ is the k-th smallest value of $\{\tau_h(v,m) : (v,m) \in S_{2,h}\}$, and $\hat{t}_h = \infty$ if $k_h > |S_{2,h}|$. The threshold value $\hat{t}_h$ is can be computed directly since the calibration set of class h only contains samples of this class. The prediction set of a new input x contains every class whose scaled envelope accepts the scores of x,

$$C(x) = \big\{ h \in \mathscr{H} : \ \tau_h\big(v_h(x), m_h(x)\big) \leq \hat{t}_h \big\}. \tag{2}$$

Theorem 1.1 (Class-conditional coverage)**.** *Assume that for every class h, the pairs (v_h, m_h) of the samples in $S_{1,h}$ and $S_{2,h}$ and of a new sample with true class h are exchangeable. Then, for every $h \in \mathscr{H}$,*

$$\mathbb{P}\big(Y \in C(X) \mid Y = h\big) \ \geq \ 1-\alpha,$$

and in particular $\mathbb{P}(Y \in C(X)) \geq 1-\alpha$. If in addition the values of τ_h are almost surely distinct, then $\mathbb{P}(Y \in C(X) \mid Y = h) \leq 1 - \alpha + 1/(|S_{2,h}|+1)$.

Proof. Let us fix a class h and write $n = |S_{2,h}|$ and $k = k_h$. We condition on $S_{1,h}$ and on the random directions of the radial method, which are independent of $S_{2,h}$ and of the new sample. Then τ_h is a fixed function. Let $T_1, \ldots, T_n$ be its values on $S_{2,h}$ and let T_0 be its value on a new sample with $Y = h$. Since $T_0, \ldots, T_n$ are exchangeable, the rank of T_0 is uniformly distributed over $\{1, \ldots, n+1\}$ when there are no ties, and ties can only make the rank of T_0 smaller. The event $\{h \in C(X)\}$ is $\{T_0 \leq T_{(k)}\}$, so

$$\mathbb{P}\big(T_0 \leq T_{(k)}\big) \ \geq \ \frac{k}{n+1} \ \geq \ 1-\alpha.$$

If $k > n$ then $\hat{t}_h = \infty$ and the probability is 1. If the values are almost surely distinct, the probability equals $k/(n+1) < 1 - \alpha + 1/(n+1)$. Averaging over $S_{1,h}$ and the directions gives the class conditional statement, and averaging over h with the class probabilities gives the marginal one. □

This is the Mondrian form of conformal prediction [1]. It does not give coverage conditional on an individual input x, which is not possible in finite samples without strong assumptions. The observed sets m_h are part of the exchangeable pairs, so a new sample must have the same missingness mechanism as the calibration samples. If a class has few calibration samples, so that $k_h > |S_{2,h}|$, its threshold is infinite and it is always accepted. The library can also return the class with the smallest normalized score $\tau_h/\hat{t}_h$ when $C(x)$ is empty. This only enlarges the set, so the lower bound still holds, but we do not use this option in our experiments, which means that empty sets count as errors.

2 Envelope constructions

Each of the three envelope construction methods follows the same two stage calibration procedure: the shape-discovery set S_1 is used to determine the geometric shape of the envelope, then the size scaling set S_2 is used to determine the scaling factor $\hat{t}$. Each construction assigns a scalar score, $\tau(v,m)$, to a score vector v with observed set m. A candidate class is then included in the prediction set when $\tau(v,m) \le \hat{t}$, where $\hat{t}$ is calculated on S_2 as in Section 1. For both the collapsed and radial methods, we define $\tau(v) = \inf\{t > 0 : v \in t \cdot E\}$ using an envelope E which is star shaped with respect to the origin and give a closed form expression for it. The strip method works in the same way, except that the single scaling factor $\hat{t}$ multiplies the limits of the staircase rather than dilating the whole region, so τ is defined directly as a maximum of ratios.

We first fix a class h and drop the index h. K denotes the number of coordinates that remain after removing the coordinates that are missing for every sample of S_1. For a vector $v \in [0,\infty)^K$ and observed set $m \subseteq \{1,\dots,K\}$, we write v_j, $j \in m$, for just its observed entries. We use $\varepsilon = 10^{-12}$ to prevent division by zero, and $Q_p(\cdot)$ denotes the empirical p-quantile.

Collapsed envelope. The basic envelope construction simply collapses the K-dimensional score vector to a scalar score by taking the mean over the observed coordinates, and calibrates a single quantile on S_1:

$$\bar{v} = \frac{1}{|m|}\sum_{j\in m} v_j, \qquad \tilde{q} = Q_{1-\alpha}\big(\{\bar{v} : v \in S_1\}\big), \qquad \tau(v,m) = \frac{\bar{v}}{\tilde{q}}, \qquad E = \{v : \bar{v} \le \tilde{q}\}. \tag{3}$$

This method discards the geometric information that the K coordinates provide. Two vectors with the same mean are treated identically whatever the distribution of their coordinates is, so a single coordinate with strong evidence (a very small v_j) is averaged together with all the others. The accepted region is a halfspace in the observed coordinates. It serves as a mean score baseline for the multidimensional constructions below. After calibration on S_2, a vector is accepted when $\bar{v}/\tilde{q} \le \hat{t}$, which is the same as $\bar{v} \le \hat{t}\,\tilde{q}$. Here $\hat{t}$ is the k-th smallest value of $\bar{v}/\tilde{q}$ over $v \in S_2$, so $\hat{t}\,\tilde{q}$ is just the k-th smallest value of of $\bar{v}$ over $v \in S_2$.

Radial envelope. The radial envelope method retains the directional information of the scores. It works by calibrating a boundary radius as a function of direction on the unit sphere, which is restricted to the positive orthant since all nonconformity scores are nonnegative. We sample M directions $\{u_r\}_{r=1}^{M}$ uniformly from the positive orthant by drawing independent vectors $g_r \sim |\mathcal{N}(0, I_K)|$ and normalizing them, $u_r = g_r/\|g_r\|$. For each vector $v \in S_1$, we compute its norm $\|v\|$ and unit direction $d_v = v/\|v\|$ using only the observed coordinates. For each direction u_r, we collect the n_r points of S_1 within an angular sector of half width

$\delta = 30^\circ$, that is, those with $d_v \cdot u_r \geq \cos\delta$. The directional radius $\tilde{q}_r$ is then calibrated as

$$\tilde{q}_r = \begin{cases} Q_{1-\alpha}\big(\{\|v\| : d_v \cdot u_r \geq \cos\delta\}\big) & \text{if } n_r \geq n_{\min}, \\ w_r\, Q_{1-\alpha}\big(\{\|v\| : d_v \cdot u_r \geq \cos\delta\}\big) + (1-w_r)\,\tilde{q}_{\text{marg}} & \text{if } 0 < n_r < n_{\min}, \\ \tilde{q}_{\text{marg}} & \text{if } n_r = 0, \end{cases} \tag{4}$$

where $n_{\min} = \max\{10, \lfloor 0.2\,|S_1| \rfloor\}$ is the minimum number of points required, $w_r = n_r/n_{\min}$, and $\tilde{q}_{\text{marg}} = Q_{1-\alpha}(\{\|s\| : s \in S_1\})$ is the marginal fallback radius. This helps us avoid a hard cutoff between sufficient and insufficient local support. To get a smooth boundary, we compute the radius in an arbitrary direction d as a weighted average of all M calibrated radii,

$$R(d) = \sum_{r=1}^{M} \omega_r(d)\,\tilde{q}_r, \qquad \omega_r(d) = \frac{\exp(\kappa\, d \cdot u_r)}{\sum_{r'=1}^{M} \exp(\kappa\, d \cdot u_{r'})}, \tag{5}$$

where $\kappa > 0$ controls the angular smoothness (a larger κ gets closer to nearest-direction assignment). We evaluate R at the directions $a_p \in \{u_1, \ldots, u_M, e_1, \ldots, e_K\}$, where the e_j are the coordinate axes. This gives us the boundary points $b_p = R(a_p)\,a_p$. The envelope E is the union of the boxes $\{v : 0 \leq v \leq b_p\}$ over all p, and the scaling function has the closed form:

$$\tau(v, m) = \min_{p} \max_{j \in m} \frac{v_j}{b_{pj} + \varepsilon}, \tag{6}$$

which is $\inf\{t > 0 : v \in t \cdot E\}$ computed on the observed coordinates. We use $M = 250$ and $\kappa = 8$ by default.

$\to \to$ (twice, in the limits equation)

Strip envelope. The strip envelope works by defining conditional bounds for each target coordinate i based on the bin occupied by another coordinate j. We partition the range of the observed S_1 values of coordinate j into NB equal width bins, and write $\beta_j(x)$ for the bin that contains x (a value below the smallest or above the largest observed S_1 value of coordinate j is put in the first or last bin, respectively). We work with the level $\alpha' = \alpha/K$, a Bonferroni type correction for the K coordinates. It only affects the shape of the envelope, since the coverage is controlled by $\hat{t}$. For each bin n containing $n_{j,n}$ points of S_1 with both v_j and v_i observed, the upper limit along coordinate $i \neq j$ is calibrated as

$$L_{j,i,n} = \begin{cases} Q_{1-\alpha'}\big(\{v_i : v \in S_1,\ \beta_j(v_j) = n\}\big) & \text{if } n_{j,n} \geq n_0, \\ w_{j,n}\, Q_{1-\alpha'}\big(\{v_i : v \in S_1,\ \beta_j(v_j) = n\}\big) + (1-w_{j,n})\,\tilde{q}_i & \text{if } 0 < n_{j,n} < n_0, \\ \tilde{q}_i & \text{if } n_{j,n} = 0, \end{cases} \tag{7}$$

where $\tilde{q}_i$ is the marginal $(1-\alpha')$ quantile of coordinate i over S_1, n_0 is the minimum number of points required in a bin, and $w_{j,n} = n_{j,n}/n_0$ shrinks the limit of underpopulated bins towards it, which is similar to the blending rule of the radial envelope. Then there is a monotonicity pass that makes sure that the limits are nonincreasing in the bin index,

$$L_{j,i,n} \leftarrow \max_{n' \geq n} L_{j,i,n'}. \tag{8}$$

This ensures that a vector with a smaller value (more confident) in the conditioning coordinate j never receives a tighter bound on coordinate i than a vector with a larger value in j. The envelope E is a coordinatewise monotone staircase, which is the intersection of these pairwise constraints. If a coordinate

is unobserved, we ignore all the pairwise constraints involving it, so it is neither used as a target nor as a conditioning coordinate. The scaling function is

$$\tau(v,m) = \begin{cases} \max\limits_{\substack{j,i\in m \\ j\neq i}} \dfrac{v_i}{L_{j,i,\beta_j(v_j)} + \varepsilon}, & |m| \geq 2, \\ \dfrac{v_j}{\tilde{q}_j + \varepsilon}, & m = \{j\}, \end{cases} \tag{9}$$

so a vector is accepted when $v_i \leq \hat{t} L_{j,i,\beta_j(v_j)}$ for every pair of observed coordinates. The single scalar $\hat{t}$ multiplies all the limits, while the bins stay fixed. We use $NB = 8$ and $n_0 = 3$ by default.

Properties and missing scores. The three scaling functions share one structural property.

Proposition 2.1. *Fix an observed set m and let v, v' be score vectors with observed set m such that $v'_j \leq v_j$ for all $j \in m$. Then $\tau(v',m) \leq \tau(v,m)$ for each of the three constructions. In particular, every accepted region $\{\tau \leq \hat{t}\}$ is downward closed, so if a candidate class is accepted, it stays accepted when its scores improve.*

Proof. For (3) this is clear, because $\tilde{q}$ does not depend on v and the mean is nondecreasing in each coordinate. In (6) each ratio $v_j/(b_{pj} + \varepsilon)$ is nondecreasing in v_j, and a maximum and a minimum of nondecreasing functions is nondecreasing. For (9), $v'_j \leq v_j$ gives $\beta_j(v'_j) \leq \beta_j(v_j)$ because β_j is nondecreasing, so $L_{j,i,\beta_j(v'_j)} \geq L_{j,i,\beta_j(v_j)}$ by (8). Together with $v'_i \leq v_i$, every ratio in (9) gets smaller. The case $|m| = 1$ is again a ratio that is nondecreasing in v_j. □

The proposition is not needed for the coverage statement of Section 1. For fixed S_1 (and fixed directions u_r, which are drawn independently of S_2 and of the new observation), each τ is a fixed function of (v,m), so the same rank argument applies to all three. Monotonicity is only a sanity property of the learned shapes.

For a vector with observed set m, the collapsed method averages over m, and the radial and strip methods only use the coordinates in m. If no coordinate is observed, the collapsed method rejects the candidate class, while the radial and strip methods accept it. During fitting we require every vector in S_1 and S_2 to have at least one observed coordinate, so the calibration does not cover this case and we make no coverage claim for it.

References

[1] Vladimir Vovk, Alexander Gammerman, and Glenn Shafer. *Algorithmic Learning in a Random World*. Springer Science & Business Media, 2005.